ICT CORNER

Recent developments in physics

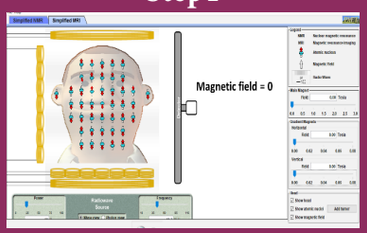
In this activity you will be able to (i) observe the changes in the nuclear spins of the hydrogen nuclei of your water molecules due to the external magnetic field (ii)find out the resonance frequency that promote a resultant photon.

Topic: MRI scan

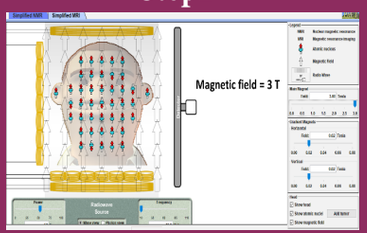
STEPS:

- Go to 'https://phet.colorado.edu/en/simulation/legacy/mri' page and download simplified MRI java file. Or go to Google → Phet → simulation → Physics → simplified MRI and download the java file.
- Open simplified MRI java file. Select simplified MRI tab.
- Observe the nuclear spins of the hydrogen nuclei present in the water molecules in brain (blue is the hydrogen nuclei). Are they aligned in same direction? What happens when you change the external magnetic field? Are they aligned in the same direction under external field? Discuss the reason.
- Now adjust the frequency bar. For a particular frequency, hydrogen nuclei emit radio waves from left to right and find out the frequency when the nuclei start broadcasting radio waves. This is resonance frequency.
- Add a tumour. Adjust the resonance frequency slightly to produce the strongest signal from the tumour. Record the tumour resonance frequency. Is there a shift?
- With the help of shift in resonance frequency, tumour inside the brain can be calculated.

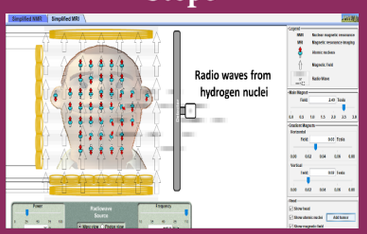
Step1



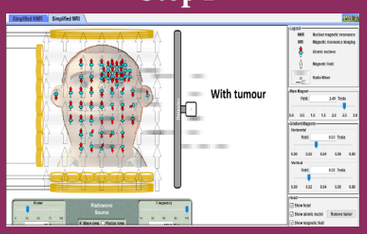
Step2



Step3



Step4



Note:

Install Java application if it is not in your browser.

URL:

https://phet.colorado.edu/en/simulation/legacy/mri

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.

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